



Embio Limited

Plant: E-21, M.I.D.C., Mahad, District Raigad, Maharashtra- 402 309, **INDIA**
Ph: +91-2145-234750 / 51/52/53/54 , **FAX** : +91-2145-232375

MATERIAL SAFETY DATA SHEET

Date : 30/07/25

PRODUCT: PSEUDOEPHEDRINE HYDROCHLORIDE

SECTION 1: IDENTIFICATION OF THE PRODUCT AND COMPANY INFORMATION

1.1 Product Identifier

Product Name: d-Pseudoephedrine Hydrochloride; (+)-Pseudoephedrine Hydrochloride

CAS Number: 345-78-8

EC Number: 206-462-1

Unique Formula Identifier: Not Applicable

1.2 Relevant identified uses of the substance or mixture and uses advised against

Chemical Family: A Phenethylamine derivative

Therapeutic Category: Adrenergic (vasoconstrictor

Nanoform: does not exhibit nanoforms

1.3 Details of the supplier of the safety data sheet · Manufacturer/Supplier:

Embio Limited

Corporate Office :501, Sentinel,

Central Avenue Road,

Hiranandani Gardens, Powai,

Mumbai- 400 076,

INDIA

Emergency Telephone No. +91-22- 66797200, +91-22-25703423

Fax No.: +91-22-66797222

Web: www.embio.co.in

Manufacturing Site : Plot E-21, MIDC Industrial Area,

Mahad, Raigad-402309, Maharashtra,

India

Emergency Telephone No. +91-8291925955-58/60/61

Fax No.: +91-2145-232375

Web: www.embio.co.in

Regd Office: 501, SENTINEL, Hiranandani Gardens, Powai, MUMBAI – 400 076, **INDIA**,
Ph: + 91-22- 6679 7200 / 25703423, **FAX** :+ 91-22-6679 7222 Email: embio@embio.co.in



Embio Limited

Plant: E-21, M.I.D.C., Mahad, District Raigad, Maharashtra- 402 309, **INDIA**
Ph: +91-2145-234750 / 51/52/53/54 , **FAX** : +91-2145-232375

SECTION 2: HAZARD IDENTIFICATION

2.1 Classification of substance or mixture

Classification according to Regulation (EC) No 1272/2008

Warning



- Acute Tox. 4 H302 Harmful if swallowed
- Skin Sense. 1 H317 May cause an allergic skin reaction

2.2 Label Elements

- Labeling according to regulation (EC) No 1272/2008
- The substance classified and labelled according to CLP regulation.
- Hazard Pictogram

Warning



-Hazard statements

- H302 Harmful if swallowed
H 317 May cause an allergic skin reaction

-Precautionary Statements

- P261 Avoid Breathing dust/ fume/ gas/ mist/ vapour/ spray
P280 Wear Protecting gloves
P301+P312 IF SWALLOWED: Call a poison center/ doctor
P333+P313 If skin irritation or rash occurs: Get medical advice/attention
P501 Dispose of contents/container in accordance with local/ regional/ national/ international regulations

2.3 Other Hazards

-Result of PBT and vPvB assessment

PBT: Not Applicable

vPvB: Not Applicable

- Endocrine disrupting properties

No data available



Embio Limited

Plant: E-21, M.I.D.C., Mahad, District Raigad, Maharashtra- 402 309, **INDIA**
Ph: +91-2145-234750 / 51/52/53/54 , **FAX** : +91-2145-232375

SECTION 3: COMPOSITION/INFORMATION OF INGREDIENTS

3.1 Chemical Characterization: Substance

- CAS Number Description: 345-78-8 (+)-Pseudoephedrine HCl
- Identification Number(s) None
- EC Number: 206-462-1
- RTECS: UL5950000
- **M-factor and the acute toxicity estimate for substances:** Not determined
- **Additional Information:** For wording of the listed hazard phrases refer section 16

SECTION 4: FIRST AID MEASURES

4.1 Description of first aid measures

- General Information

Symptoms of poisoning may occur even after several hours; therefore medical observation for at least 48 hours after the accident is recommended.

- After Inhalation

Supply fresh air and contact doctor

In case of unconsciousness place patient in recovery position for transport.

- After Skin contact

Immediately wash with water and soap and rinse thoroughly.

- After eye contact

Rinse open eye for several minutes under running water

- After Swallowing:

Rinse mouth. Do not induce vomiting.

Seek medical treatment.

4.2 Most important symptoms and effects, both acute and delayed

No further relevant information available

4.3 Indication of any immediate medical attention and special treatment needed

No further relevant information available.

SECTION 5: FIRE FIGHTING MEASURES

5.1 Extinguisher Media

Water spray, dry chemical, carbon dioxide or foam as appropriate for surrounding fire and materials.



Embio Limited

Plant: E-21, M.I.D.C., Mahad, District Raigad, Maharashtra- 402 309, **INDIA**
Ph: +91-2145-234750 / 51/52/53/54 , **FAX** : +91-2145-232375

5.2 Special Hazards arising from substance of mixture

Formation of toxic gases possible during heating or in case of fire.

5.3 Advice for Fire fighter

Firefighters should use self-contained breathing equipment and protective clothing.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1 Personal precautions, protective equipment and emergency procedures

Wear protective clothing.

6.2 Environmental Precautions

Do not allow to enter sewers/ surface or ground water.

6.3 Methods and material for containment and cleaning up

Pickup mechanically.

6.4 Reference to other sections

See section 7, for information on safe handling

See section 8, for information on personal protection equipment

See section 13, for disposal information

SECTION 7: HANDLING AND STORAGE

7.1 Precautions for safe handling

Store in cool and dry place in tightly closed receptacles.

- **Information about fire and explosion protection:** No special measures required

7.2 Condition for safe storage, including any incompatibilities

- Storage

Store only in original receptacles unless otherwise mentioned in COA.

Keep container in a well-ventilated place. Keep away from source of ignition and heat.

- **Information about storage in one common storage facility**

Store away from food stuff.

- **Further information about storage conditions**

None

7.3 Specific end use(s)

No further relevant information available

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

- **Additional information about design of technical facilities:** No further data; see item 7.

8.1 Control parameters

- **Ingredient with limit values that require monitoring at work place:** Not required

- **Additional information:** Lists used were valid at the time of SDS preparation.

Regd Office: 501, SENTINEL, Hiranandani Gardens, Powai, MUMBAI – 400 076, **INDIA**,
Ph: + 91-22- 6679 7200 / 25703423, **FAX** :+ 91-22-6679 7222 Email: embio@embio.co.in



Embio Limited

Plant: E-21, M.I.D.C., Mahad, District Raigad, Maharashtra- 402 309, **INDIA**
Ph: +91-2145-234750 / 51/52/53/54 , **FAX** : +91-2145-232375

8.2 Exposure Controls

- **General protective and hygienic measures:** Immediately remove all soiled and contaminated clothing. Wash hand in the breaks and at the end of the work. Use Personal protective equipment

- **Respiratory Protection:** Not required.

Use suitable respiratory protective device in case of insufficient ventilation.

- **Protection of Hands:**

The glove material has to be impermeable and resistant to the product/ the substance/ the preparation. Selection of the glove material on consideration of the penetration times, rates of diffusion and the degradation.

The protective gloves to be used must comply with the specification of EC directive 89/686/EEC and the related standard EN374



Protective gloves

- **Material of gloves:** Nitrile rubber, NBR

- **Penetration time of glove material:** The exact break through time has to be found out by the manufacturer of the protective gloves and has to be observed.

- **Eye Protection:** Safety glasses

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1 Information on basic physical and chemical properties

- Form	:	Solid
- Colour	:	White
- Odour	:	Odourless or faint characteristic odour
- Odour threshold	:	Not determined
- Melting point	:	182-186°C
- Solubility		
Water	:	Freely soluble
- Boiling point	:	Not determined
- Evaporation rate	:	Not applicable
- Specific gravity	:	Not applicable
- Reactivity in water	:	Not reactive
- Vapor Pressure	:	Not applicable
- Density	:	Not determined
- Flash Point	:	Not applicable
- Flamability	:	Not determined

Regd Office: 501, SENTINEL, Hiranandani Gardens, Powai, MUMBAI – 400 076, **INDIA**,
Ph: + 91-22- 6679 7200 / 25703423, **FAX** :+ 91-22-6679 7222 Email: embio@embio.co.in



Embio Limited

Plant: E-21, M.I.D.C., Mahad, District Raigad, Maharashtra- 402 309, **INDIA**
Ph: +91-2145-234750 / 51/52/53/54 , **FAX** : +91-2145-232375

- Ignition temperature	:	Not determined
- Decomposition temperature	:	Not determined
- Auto Ignition temperature	:	Not determined
- Explosive properties	:	Not determined
- Explosion limits		
Lower	:	Not determined
Upper	:	Not determined
- Partition coefficient		
Octanol: water	:	Not determined

9.2 Other information: Not further relevant information available

SECTION 10: STABILITY AND REACTIVITY

10.1 Reactivity

Stable under normal conditions.

10.2 Chemical Stability

Stable under normal condition

- Thermal decomposition/ conditions to be avoided:

Formation of toxic gases is possible during heating or in case of fire.

10.3 Possibility of hazardous reaction

No dangerous reaction known

10.4 Condition to avoid

Heat.

10.5 Materials to Avoid

Oxidizing agents.

10.6 Hazardous decomposition of products

Formation of toxic gases (Carbon monoxide, Carbon dioxide, Nitrogen oxides, Hydrogen chloride gas) is possible during heating or in case of fire.

SECTION 11: TOXICOLOGICAL INFORMATION

11.1 Information on toxicological effects

- **Acute toxicity:** Harmful if swallowed



Embio Limited

Plant: E-21, M.I.D.C., Mahad, District Raigad, Maharashtra- 402 309, **INDIA**
Ph: +91-2145-234750 / 51/52/53/54 , **FAX** : +91-2145-232375

LD/LC50 values relevant for classification		
Oral	LD50	371mg/kg (mouse)

- Primary irritant effect

Skin corrosion/ irritation : Based on available data, the classification criteria not met
Serious eye damage/ irritation : Based on available data, the classification criteria not met
Respiratory/ skin sensation : May cause allergic skin reaction

- CMR effects (carcinogenicity, mutagenicity and toxicity for reproduction)

Germ cell mutagenicity : Based on available data, the classification criteria are not met.
Carcinogenicity : Based on available data, the classification criteria are not met.
Reproductive toxicity : Based on available data, the classification criteria are not met.
STOT-single exposure : Based on available data, the classification criteria are not met.
STOT-repeated exposure : Based on available data, the classification criteria are not met.

Aspiration hazard : Based on available data, the classification criteria are not met.

- Signs and Symptoms of exposure

Adverse reactions: nausea, vomiting, constipation, dizziness, sedation, palpitations, pruritus, anxiety, tenseness, restlessness, tremor, weakness, pallor, respiratory difficulty, dysuria, insomnia, hallucinations, convulsions, CNS depression, arrhythmias, cardiovascular collapse with hypotension. Individuals with preexisting health conditions of hypertension, diabetes mellitus, ischemic heart disease, hyperthyroidism, increased intraocular pressure or prostatic hypertrophy should exert extra care when working with sympathomimetics.

- **Target organ information:** Nerves. Heart.

SECTION 12: ECOLOGICAL INFORMATION

12.1 Toxicity

- **Aquatic toxicity:** No further relevant information available

12.2 Persistence and degradability: No further relevant information available

12.3 Bioaccumulative potential: No further relevant information available.

12.4 Mobility in soil: No further relevant information available.

- **Additional ecological information:**

- **General notes:**

Water hazard class 1 (German Regulation) (Assessment by list): slightly hazardous for water
Do not allow undiluted product to reach ground water, water course or sewage system.

12.5 Results of PBT and vPvB assessment

- **PBT:** Not applicable.

- **vPvB:** Not applicable.

Regd Office: 501, SENTINEL, Hiranandani Gardens, Powai, MUMBAI – 400 076, **INDIA**,
Ph: + 91-22- 6679 7200 / 25703423, **FAX** :+ 91-22-6679 7222 Email: embio@embio.co.in

	Embio Limited Plant: E-21, M.I.D.C., Mahad, District Raigad, Maharashtra- 402 309, INDIA Ph: +91-2145-234750 / 51/52/53/54 , FAX : +91-2145-232375
---	--

12.6 Other adverse effects: No further relevant information available. No evidence of endocrine disrupting properties.

SECTION 13: DISPOSAL CONSIDERATIONS

13.1 Waste treatment methods

Recommendation

Must not be disposed of together with household garbage. Do not allow product to reach sewage system.

European waste catalogue

Waste disposal key numbers from EWC have to be assigned depending on origin and processing.

Uncleaned packaging:

Recommendation: Dispose of in accordance with national regulations.

Recommended cleansing agents: Water, if necessary together with cleansing agents.

SECTION - 14 TRANSPORT INFORMATION

14.1 UN-Number

- ADR, ADN, IMDG, IATA Not applicable
- ADR, ADN, IMDG, IATA Not applicable

14.2 Transport hazard class(es)

- ADR, ADN, IMDG, IATA
- Class Not applicable

14.3 Packing group

- ADR, IMDG, IATA Not applicable

14.4 Environmental hazards: Not applicable.

14.5 Special precautions for user: Not applicable.

14.6 Transport in bulk according to Annex II of Marpol and the IBC Code: Not applicable.

-UN "Model Regulation": Not applicable



Embio Limited

Plant: E-21, M.I.D.C., Mahad, District Raigad, Maharashtra- 402 309, **INDIA**
Ph: +91-2145-234750 / 51/52/53/54 , **FAX** : +91-2145-232375

SECTION - 15 REGULATORY INFORMATION

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

- **Directive 2012/18/EU** ·
- **Named dangerous substances - ANNEX I Substance is not listed.**
- **National regulations:**

Other regulations, limitations and prohibitive regulations

The substance is considered a controlled drug in some countries and needs to be treated according to the applicable regulations.

15.2 **Chemical safety assessment:** A Chemical Safety Assessment has not been carried out.

SECTION - 16 OTHER INFORMATION

The information in this safety data sheet (SDS) has been prepared with due care and is true and accurate to the best of our knowledge. The user must determine the suitability of the information for its particular purpose, ensure compliance with existing laws and regulations, and be aware that other or additional safety or performance considerations may arise when using, handling and/ or storing the material. The information in this SDS does not purport to be all inclusive or a guarantee as to the properties of the material supplied, and should be used only as a guide. LGC makes no warranties or representations as to the accuracy and completeness of the information contained herein, shall not be held responsible for the suitability of this information for the user's intended purposes or the consequences of such use, and shall not be liable for any damage or loss, howsoever arising, direct or otherwise.

- Abbreviations and acronyms:

ADR: Accord européen sur le transport des marchandises dangereuses par Route (European Agreement concerning the International

Carriage of Dangerous Goods by Road)

IMDG: International Maritime Code for Dangerous Goods

IATA: International Air Transport Association

GHS: Globally Harmonised System of Classification and Labelling of Chemicals

EINECS: European Inventory of Existing Commercial Chemical Substances

CAS: Chemical Abstracts Service (division of the American Chemical Society)

LC50: Lethal concentration, 50 percent

LD50: Lethal dose, 50 percent

PBT: Persistent, Bio accumulative and Toxic

vPvB: very Persistent and very Bio accumulative

Regd Office: 501, SENTINEL, Hiranandani Gardens, Powai, MUMBAI – 400 076, **INDIA**,
Ph: + 91-22- 6679 7200 / 25703423, **FAX** :+ 91-22-6679 7222 Email: embio@embio.co.in



Embio Limited

Plant: E-21, M.I.D.C., Mahad, District Raigad, Maharashtra- 402 309, **INDIA**
Ph: +91-2145-234750 / 51/52/53/54 , **FAX** : +91-2145-232375

Acute Tox. 4: Acute toxicity – Category 4

Skin Sens. 1: Skin sensitisation – Category 1

- Sources

Tables 3.1 and 3.2 from Annex 6 of EC 1272/2008, EC 1907/2006, EH40/2005 as amended 2011, Registry of Toxic Effects of Chemical Substances (RTECS), The Dictionary of Substances and their Effects, 1st Edition, IUCLID.

- **Data compared to the previous version altered.** All sections have been updated.

ORIGINAL CONTROLLED